

▼ The Sugar Trap Market Gap Analysis

▼ PART 1: Dataset Loading & Libraries

▼ 1.Useful Libraries to be used:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import re
from collections import Counter
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
```

▼ 2.Loading and Analyzing Open Food Fact dataset

```
# Getting dataset Link
URL = "https://static.openfoodfacts.org/data/en.openfoodfacts.org.products.csv.gz"

# Loading Dataset as "df"
df = pd.read_csv(URL, sep='\t', nrows=500_000, low_memory=False)

# Allow the notebook to view all columns
pd.set_option('display.max_columns', None)
```

```
# View first 5 of the food dataset
df.head()
```

	code	url	creator	created_t	created_datetime	last_modified_t	last_modified_datetime	last_moo
0	54	http://world-en.openfoodfacts.org/product/0000...	kiliweb	1582569031	2020-02-24T18:30:31Z	1733085204	2024-12-01T20:33:24Z	
1	63	http://world-en.openfoodfacts.org/product/0000...	kiliweb	1673620307	2023-01-13T14:31:47Z	1750061386	2025-06-16T08:09:46Z	bc
2	114	http://world-en.openfoodfacts.org/product/0000...	kiliweb	1580066482	2020-01-26T19:21:22Z	1751035658	2025-06-27T14:47:38Z	
3	431	http://world-en.openfoodfacts.org/product/0000...	kiliweb	1714301712	2024-04-28T10:55:12Z	1714301721	2024-04-28T10:55:21Z	
4	105	http://world-en.openfoodfacts.org/product/0000...	kiliweb	1572117743	2019-10-26T19:22:23Z	1738073570	2025-01-28T14:12:50Z	

```
print(f"The dataset consist of {df.shape[0]} rows and {df.shape[1]} columns.")
```

```
The dataset consist of 500000 rows and 211 columns.
```

```
# Further information about the dataset
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500000 entries, 0 to 499999
Columns: 211 entries, code to water_100g
dtypes: float64(136), int64(3), object(72)
memory usage: 804.9+ MB
```

The Food Fact Dataset has 211 columns and has been set to use first 500,000 rows for the moment.

▼ PART 2: User Stories & Acceptance Criteria

▼ Story 1: Data Ingestion & "The Clean Up"

```
# Selecting few columns to use, dataframe named "food_df"
food_df = df[['product_name', 'categories_tags', 'sugars_100g',
              'proteins_100g', 'fat_100g', 'fiber_100g', 'ingredients_text']]

food_df.head()
```

	product_name	categories_tags	sugars_100g	proteins_100g	fat_100g	fiber_100g	ingredients_text	
0	Limonade artisanale a la rose	NaN	NaN	NaN	NaN	NaN	NaN	
1	M&M white	NaN	NaN	NaN	NaN	NaN	Weizenmehl, Rapsöl, Speisesalz, 1,7% Meersalz,...	
2	Chocolate n3	NaN	NaN	NaN	NaN	NaN	NaN	
3	Pâte de fruits	NaN	NaN	NaN	NaN	NaN	NaN	
4	Paleta oran reserva	en:beverages-and-	...	...	...	...	Thiamin, Biotin, Chromium,	

As it is visible in the dataframe we have so many missing values, next step is to delete rows with missing values in empty "sugars_100g", "proteins_100g", or "product_name".

```
# I drop rows based on 3 critical columns
food_df = food_df.dropna(subset=['product_name', 'sugars_100g',
                                'proteins_100g'])

# Shape of dataframe after removing some rows
food_df.shape

(109826, 7)
```

▼ Handling Outliers (nutrients > 100g/100g product)

```
# Remove Biologically impossible values among 4 nutrients columns
nutrients_columns = ['sugars_100g', 'proteins_100g', 'fat_100g', 'fiber_100g']

for column in nutrients_columns:
    food_df = food_df[(food_df[column] >= 0) & (food_df[column] <= 100)]

# Cleaned dataset now
print(f"Dataset now has: {len(food_df):,} rows after cleaning")

Dataset now has: 80,707 rows after cleaning
```

```
food_df.head()
```

	product_name	categories_tags	sugars_100g	proteins_100g	fat_100g	fiber_100g	
689	Pinto Bean	en:asian-style-ready-meal	4.900000	17.500000	10.200000	3.800000	
691	Keto & GF Granola	NaN	3.225806	19.677419	54.838710	9.677419	
835	Thai Red Curry (produit préparé)	NaN	1.200000	1.400000	0.600000	0.700000	
840	pasta	en:beverages-and-beverages-preparations,en:bev...	1.900000	6.700000	6.400000	0.800000	
848	Dunnes Stores Irish Beef & Pork Quarter Pounders	NaN	0.521739	15.652174	11.304348	0.782609	

Next steps:

[Generate code with food_df](#)[New interactive sheet](#)

Story 2: The Category Wrangler

```
# Assign food products in categories based on keywords found in 'categories_tags'
def assign_category(tags):
    # If 'categories_tags' is missing (NaN), assign 'Other' category
    if pd.isna(tags):
        return 'Other'

    # Convert tags to lowercase for case-insensitive matching
    tags = tags.lower()

    # Check for keywords related to 'Protein Bars' category
    if any(k in tags for k in [
        'protein-bar', 'energy-bar', 'granola-bar', 'sports-nutrition',
        'protein-powder', 'meal-replacement', 'nutrition-bar'
    ]): return 'Protein Bars & Supplements'

    # Check for keywords related to 'Confectionery' category
    if any(k in tags for k in [
        'chocolate', 'candy', 'confection', 'sweet', 'gummy',
        'marshmallow', 'caramel', 'nougat', 'lollipop', 'fudge'
    ]): return 'Confectionery'

    # Check for keywords related to 'Dairy' category
    if any(k in tags for k in [
        'dairy', 'milk', 'yogurt', 'yoghurt', 'cheese', 'butter',
        'cream', 'kefir', 'whey', 'casein', 'egg'
    ]): return 'Dairy & Eggs'

    # Check for keywords related to 'Cereals' category
    if any(k in tags for k in [
        'cereal', 'oat', 'muesli', 'porridge', 'granola',
        'rice', 'wheat', 'grain', 'quinoa', 'barley', 'corn-flour',
        'pasta', 'noodle', 'bread', 'flour'
    ]): return 'Cereals & Grains'

    # Check for keywords related to 'Beverages' category
    if any(k in tags for k in [
        'beverage', 'drink', 'juice', 'soda', 'water', 'tea',
        'coffee', 'smoothie', 'shake', 'lemonade', 'beer', 'wine',
        'spirits', 'alcohol', 'en:beverages'
    ]): return 'Beverages'

    # Check for keywords related to 'Bakery - Snacks' category
    if any(k in tags for k in [
        'biscuit', 'cookie', 'cake', 'pastry', 'muffin', 'wafer',
        'cracker', 'pretzel', 'chip', 'crisp', 'popcorn', 'snack',
        'puff', 'tortilla', 'en:biscuits', 'en:crackers'
    ]): return 'Bakery & Snacks'

    # Check for keywords related to 'Legumes - Plant proteins' category
    if any(k in tags for k in [
        'legume', 'lentil', 'bean', 'pea', 'chickpea', 'soy',
        'tofu', 'tempeh', 'edamame', 'hummus', 'pulse',
        'en:legumes', 'en:beans', 'en:lentils'
    ]): return 'Legumes & Plant Protein'

    # Check for keywords related to 'Nuts - Seeds' category
```

```
if any(k in tags for k in [
    'nut', 'almond', 'cashew', 'walnut', 'peanut', 'seed',
    'sunflower', 'pumpkin-seed', 'chia', 'flax', 'hemp',
    'en:nuts', 'en:seeds', 'nut-butter'
]): return 'Nuts & Seeds'

# Check for keywords related to 'Fruits - Vegetables' category
if any(k in tags for k in [
    'fruit', 'vegetable', 'veggie', 'produce', 'dried-fruit',
    'raisin', 'berry', 'apple', 'banana', 'broccoli', 'spinach',
    'en:fruits', 'en:vegetables', 'en:dried-fruits'
]): return 'Fruits & Vegetables'

# Check for keywords related to 'Condiments - Seasonings' category
if any(k in tags for k in [
    'condiment', 'sauce', 'seasoning', 'spice', 'herb', 'salt',
    'pepper', 'yeast', 'vinegar', 'dressing', 'marinade',
    'en:condiments', 'en:spices', 'en:seasonings'
]): return 'Condiments & Seasonings'

# Check for keywords related to 'Meat - Fish' category
if any(k in tags for k in [
    'meat', 'chicken', 'beef', 'pork', 'fish', 'seafood',
    'salmon', 'tuna', 'turkey', 'jerky', 'en:meats', 'en:fish'
]): return 'Meat & Fish'

# If no specific category matches, assign 'Other'
return 'Other'

# Apply the 'assign_category' function to the 'categories_tags' column
# and store the results in a new column called 'primary_category' in the cleaned 'food_df' DataFrame.
food_df['primary_category'] = food_df['categories_tags'].apply(assign_category)

# coverage
food_df['primary_category'].value_counts()
```

	count
primary_category	
Other	41740
Cereals & Grains	12477
Dairy & Eggs	7265
Confectionery	6275
Beverages	6157
Meat & Fish	2310
Bakery & Snacks	2076
Condiments & Seasonings	1922
Protein Bars & Supplements	214
Legumes & Plant Protein	127
Fruits & Vegetables	113
Nuts & Seeds	31

dtype: int64

Double-click (or enter) to edit

▼ Story 3: The "Nutrient Matrix" Visualization

```
#Only needs category + sugar + protein + quadrant
def label_quadrant(row):
    if row['sugars_100g'] < 10 and row['proteins_100g'] > 15:
        return 'Blue Ocean – High Protein Low Sugar'
    elif row['sugars_100g'] >= 10 and row['proteins_100g'] > 15:
        return 'High Protein High Sugar'
    elif row['sugars_100g'] < 10 and row['proteins_100g'] <= 15:
        return 'Low Sugar Low Protein'
```

```

else:
    return 'Danger Zone – High Sugar Low Protein'

food_df['quadrant'] = food_df.apply(label_quadrant, axis=1)
print("quadrant added")

# Export for PowerBI 20K samples for fast process
scatter_df = food_df[[
    'product_name',
    'primary_category',
    'sugars_100g',
    'proteins_100g',
    'quadrant'
]].dropna().sample(20_000, random_state=42)

scatter_df.to_csv('nutrient_matrix.csv', index=False)
print(f"nutrient_matrix.csv exported : {scatter_df.shape[0]:,} rows")
print(f"\nQuadrant distribution:\n{scatter_df['quadrant'].value_counts()}")

```

```

quadrant added
nutrient_matrix.csv exported : 20,000 rows

Quadrant distribution:
quadrant
Low Sugar Low Protein          10837
Danger Zone – High Sugar Low Protein    6069
Blue Ocean – High Protein Low Sugar    2732
High Protein High Sugar           362
Name: count, dtype: int64

```

✓ Story 4: The Recommendation

```

# Blue Ocean products, excluding 'Other' category
blue_ocean = food_df[
    (food_df['sugars_100g'] < 10) &
    (food_df['proteins_100g'] > 15) &
    (food_df['primary_category'] != 'Other')
].copy()

print(f"Blue Ocean products (excluding Other): {len(blue_ocean):,}")
print(f"\nBy category:\n{blue_ocean['primary_category'].value_counts()}")

```

```

Blue Ocean products (excluding Other): 4,349

By category:
primary_category
Meat & Fish          1402
Dairy & Eggs         1314
Beverages            750
Cereals & Grains     398
Bakery & Snacks       198
Protein Bars & Supplements  133
Confectionery        77
Condiments & Seasonings  43
Nuts & Seeds          17
Legumes & Plant Protein  16
Fruits & Vegetables    1
Name: count, dtype: int64

```

```

# Best category by count
best_category = blue_ocean['primary_category'].value_counts().index[0]

# Target values from that category only
category_products = blue_ocean[blue_ocean['primary_category'] == best_category]

target_protein = round(category_products['proteins_100g'].median(), 1)
target_sugar = round(category_products['sugars_100g'].quantile(0.90), 1)

print(f"Best category : {best_category}")
print(f"Target protein : {target_protein}g")
print(f"Max sugar : {target_sugar}g")
print(f"Products found : {len(category_products):,}")

Best category : Meat & Fish
Target protein : 20.0g
Max sugar : 3.4g
Products found : 1,402

```

```

insight = (
    f"Based on the data, the biggest market opportunity is in \n"
    f"**{best_category}**, specifically targeting products with \n"
    f"{target_protein}g of protein and less than {target_sugar}g of sugar."
)

print("=" * 65)
print("SUMMARY")
print("=" * 65)
print(insight)
print("=" * 65)

```

```

=====
SUMMARY
=====
Based on the data, the biggest market opportunity is in
**Meat & Fish**, specifically targeting products with
20.0g of protein and less than 3.4g of sugar.
=====

```

5. Bonus User Story: The "Hidden Gem"

```

# High Protein cluster is same as Blue Ocean where protein > 15g, sugar < 10g
# but here we focus only on ingredients_text
high_protein = food_df[
    (food_df['proteins_100g'] > 15) &
    (food_df['primary_category'] != 'Other')
].copy()

# Drop rows where ingredients_text is missing
high_protein = high_protein.dropna(subset=['ingredients_text'])

print(f"High protein products with ingredients: {len(high_protein):,}")
print(f"\nSample ingredients text:")
print(high_protein['ingredients_text'].iloc[0])

```

High protein products with ingredients: 3,850

Sample ingredients text:
Soya protein isolate (40 %), fructose, emulsifier (soya lecithin), polydextrose, oat fiber, inulin, coffee powder, soya oil, thi

```

# Known protein keywords to scan
protein_keywords = [
    'whey', 'casein', 'soy', 'soya', 'pea protein',
    'peanut', 'almond', 'egg', 'chicken', 'beef',
    'milk protein', 'rice protein', 'hemp protein',
    'salmon', 'tuna', 'lentil', 'chickpea',
    'quinoa', 'black bean', 'sunflower protein'
]

```

```

ingredient_counts = Counter()

for text in high_protein['ingredients_text'].str.lower():
    for keyword in protein_keywords:
        if keyword in text:
            ingredient_counts[keyword] += 1

# Convert to DataFrame
ingredient_df = pd.DataFrame(
    ingredient_counts.most_common(),
    columns=['protein_source', 'product_count']
)

ingredient_df['percentage'] = (
    (ingredient_df['product_count'] / len(high_protein)) * 100
).round(1)

print("=" * 50)
print("TOP PROTEIN SOURCES IN HIGH PROTEIN PRODUCTS")
print("=" * 50)
print(ingredient_df.head(10).to_string(index=False))

```

```

=====
TOP PROTEIN SOURCES IN HIGH PROTEIN PRODUCTS
=====

```

```
=====
protein_source  product_count  percentage
soy              791           20.5
peanut           551           14.3
chicken          286           7.4
almond           270           7.0
beef             244           6.3
whey             198           5.1
soya             103           2.7
egg              103           2.7
milk protein     82            2.1
pea protein      80            2.1
```

```
top_3 = ingredient_df.head(3)

print("\n" + "=" * 50)
print("TOP 3 PROTEIN SOURCES")
print("=" * 50)

for i, row in top_3.iterrows():
    print(f" #{i+1} {row['protein_source'].title():<20} "
          f"found in {row['product_count']:,} products "
          f"{row['percentage']:%}")

print("=" * 50)
```

```
=====
TOP 3 PROTEIN SOURCES
=====
#1 Soy              found in 791 products (20.5%)
#2 Peanut           found in 551 products (14.3%)
#3 Chicken          found in 286 products (7.4%)
=====
```

```
figure, ax = plt.subplots(figsize=(10, 5))

bars = ax.barh(
    ingredient_df.head(10)['protein_source'].str.title(),
    ingredient_df.head(10)['product_count'],
    color='steelblue',
    edgecolor='white'
)

# Add value labels on bars
for bar, val in zip(bars, ingredient_df.head(10)['product_count']):
    ax.text(bar.get_width() + 10, bar.get_y() + bar.get_height()/2,
            f'{val:,}', va='center', fontsize=10)

ax.set_xlabel('Number of Products')
ax.set_title('Top Protein Sources in High Protein Products', fontsize=13)
ax.invert_yaxis()
plt.tight_layout()
plt.savefig('top_protein_sources.png', dpi=150, bbox_inches='tight')
plt.show()

print("Chart saved as top_protein_sources.png")
```

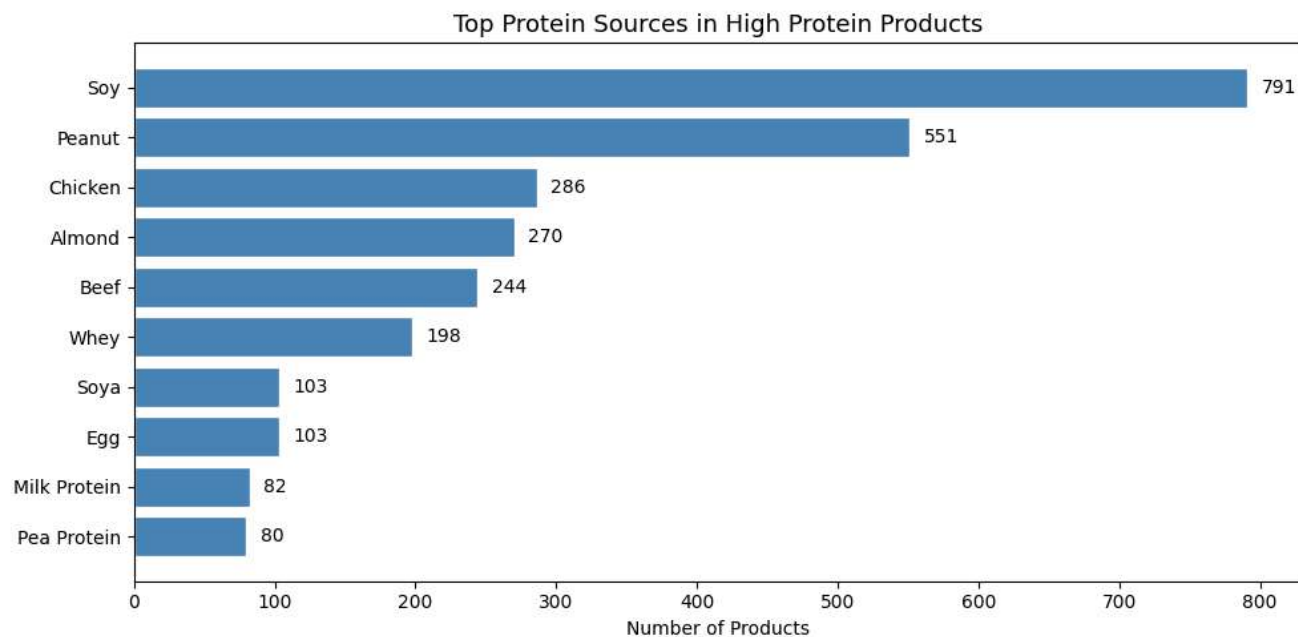


Chart saved as top_protein_sources.png

```
def health_score(row):
    """
    Composite score 0-100 based on WHO dietary guidelines:
    - Rewards: high protein, high fiber
    - Penalizes: high sugar, high fat
    """
    sugar_pen = min(row['sugars_100g'] / 10, 5)
    fat_pen = min(row['fat_100g'] / 15, 5)
    protein_rew = min(row['proteins_100g'] / 5, 5)
    fiber_rew = min(row['fiber_100g'] / 3, 5)

    raw = (protein_rew + fiber_rew - sugar_pen - fat_pen)
    return int(round((raw + 10) * 5))

food_df['health_score'] = food_df.apply(health_score, axis=1)

print("Health Score stats:")
print(food_df['health_score'].describe().round(2))
```

```
Health Score stats:
count    80707.00
mean      53.07
std       13.34
min        3.00
25%       46.00
50%       54.00
75%       61.00
max       100.00
Name: health_score, dtype: float64
```

```
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

features = ['sugars_100g', 'proteins_100g', 'fat_100g', 'fiber_100g']
cluster_df = food_df[features].dropna().copy()

scaler = StandardScaler()
X_scaled = scaler.fit_transform(cluster_df)

kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
cluster_df['cluster'] = kmeans.fit_predict(X_scaled)

df_clustered = df.loc[cluster_df.index].copy()
df_clustered['cluster'] = cluster_df['cluster']

#bring health_score in from food_df
df_clustered['health_score'] = food_df.loc[cluster_df.index, 'health_score']

#Confirm columns exist before groupby
```



```
print("Columns available:", df_clustered.columns.tolist())
```

```
cluster_profiles = df_clustered.groupby('cluster').agg(
    avg_sugar = ('sugars_100g', 'mean'),
    avg_protein = ('proteins_100g', 'mean'),
    avg_fiber = ('fiber_100g', 'mean'),
    avg_fat = ('fat_100g', 'mean'),
    avg_health = ('health_score', 'mean'),
    count = ('product_name', 'count')
).round(2)
```

```
print("\nCluster Profiles:")
print(cluster_profiles.sort_values('avg_health', ascending=False))
```

Columns available: ['code', 'url', 'creator', 'created_t', 'created_datetime', 'last_modified_t', 'last_modified_datetime', 'las

Cluster Profiles:

cluster	avg_sugar	avg_protein	avg_fiber	avg_fat	avg_health	count
0	8.73	16.74	29.13	11.27	81.84	1515
1	4.72	20.37	3.44	34.11	58.59	12771
3	5.47	6.66	2.14	5.38	55.69	53076
2	45.38	4.71	2.75	14.80	34.06	13345

```
sorted_clusters = (
    cluster_profiles['avg_health']
    .sort_values(ascending=False)
    .index.tolist()
)
```

```
tier_map = {
    sorted_clusters[0]: 'Tier 1 - Optimal',
    sorted_clusters[1]: 'Tier 2 - Good',
    sorted_clusters[2]: 'Tier 3 - Moderate',
    sorted_clusters[3]: 'Tier 4 - Poor'
}
```

```
df_clustered['health_tier'] = df_clustered['cluster'].map(tier_map)
```

```
print("Health Tier distribution:")
print(df_clustered['health_tier'].value_counts())
```

```
print("\nTier mapping confirmed:")
for cluster, tier in tier_map.items():
    row = cluster_profiles.loc[cluster]
    print(f" {tier:<25} : cluster {cluster} "
          f"| health score: {row['avg_health']} "
          f"| protein: {row['avg_protein']}g "
          f"| sugar: {row['avg_sugar']}g "
          f"| count: {int(row['count']):,}")
```

Health Tier distribution:

```
health_tier
Tier 3 - Moderate    53076
Tier 4 - Poor       13345
Tier 2 - Good       12771
Tier 1 - Optimal     1515
Name: count, dtype: int64
```

Tier mapping confirmed:

Tier 1 - Optimal	: cluster 0 health score: 81.84 protein: 16.74g sugar: 8.73g count: 1,515
Tier 2 - Good	: cluster 1 health score: 58.59 protein: 20.37g sugar: 4.72g count: 12,771
Tier 3 - Moderate	: cluster 3 health score: 55.69 protein: 6.66g sugar: 5.47g count: 53,076
Tier 4 - Poor	: cluster 2 health score: 34.06 protein: 4.71g sugar: 45.38g count: 13,345

```
colors = {
    'Tier 1 - Optimal' : '#2E7D32',
    'Tier 2 - Good'    : '#66BB6A',
    'Tier 3 - Moderate' : '#FFA726',
    'Tier 4 - Poor'     : '#EF5350'
}
```

```
fig, axes = plt.subplots(1, 2, figsize=(14, 6))
```

```
#Plot 1: Scatter colored by health tier
sample = df_clustered.sample(8000, random_state=42)
```

```
for tier, color in colors.items():
```

```

mask = sample['health_tier'] == tier
axes[0].scatter(
    sample.loc[mask, 'sugars_100g'],
    sample.loc[mask, 'proteins_100g'],
    c=color, alpha=0.4, s=8, label=tier
)

axes[0].axvline(x=10, color='green', linestyle='--', linewidth=1, alpha=0.7)
axes[0].axhline(y=15, color='green', linestyle='--', linewidth=1, alpha=0.7)
axes[0].set_xlabel('Sugar (g per 100g)')
axes[0].set_ylabel('Protein (g per 100g)')
axes[0].set_title('Products by Health Tier')
axes[0].legend(loc='upper right', fontsize=8)

# Add primary_category to df_clustered
df_clustered['primary_category'] = food_df.loc[cluster_df.index, 'primary_category']

#Plot 2: Average health score by category
cat_scores = (
    df_clustered[df_clustered['primary_category'] != 'Other']
    .groupby('primary_category')['health_score']
    .mean()
    .sort_values(ascending=True)
)

bars = axes[1].barh(
    cat_scores.index,
    cat_scores.values,
    color=[ '#2E7D32' if v >= 50 else '#FFA726' if v >= 35 else '#EF5350'
           for v in cat_scores.values]
)

axes[1].axvline(x=50, color='gray', linestyle='--', linewidth=1, alpha=0.7)
axes[1].set_xlabel('Average Health Score (0-100)')
axes[1].set_title('Average Health Score by Category')

for bar, val in zip(bars, cat_scores.values):
    axes[1].text(bar.get_width() + 0.5,
                 bar.get_y() + bar.get_height()/2,
                 f'{val:.1f}', va='center', fontsize=9)

plt.suptitle("Nutritional Health Scoring & ML Clustering",
             fontsize=13, fontweight='bold', y=1.02)
plt.tight_layout()
plt.savefig('health_clustering.png', dpi=150, bbox_inches='tight')
plt.show()

print("Chart saved as health_clustering.png")

```

Nutritional Health Scoring & ML Clustering

